

POST - SCRIPT

Science is a kind of language game. It represents a system of organized and empirical knowledge of the world. Its practice is defined by a set of standard procedures and conventions under an assumption known as the scientific method. This method, in essence, defines an inductive, spiral process of data collection, hypothesis formation, theory and model construction and prediction generation. The practitioners of science can devise tentative solutions to various situations, experiment with them and modify or discard them in the light of the observation conducted.

The pursuit of any discipline of science is a social activity initiated and sustained by a community of scientists. In general, the definition of the discursive domain of any scientific discipline is bound by paradigm(s) embodying sets of beliefs, rules and practices agreed upon by its participants. These paradigms are regulated by the mode of production and the ideological superstructure peculiar to that time. These paradigms reflect the Weltanschauung or world view of that period. Any breach of the rules defined by the dominant paradigms may result in ostracization or persecution. In a nutshell, these paradigms may serve as a form of social and ideological control restraining the activities of the participants.

The major paradigm of any scientific discipline forms the core of that discipline. It is held by the majority of its practitioners with much tenacity. Sometimes, it may be hailed as a kind of absolute, inviolable dogma. The periphery of a scientific discipline, however, contains certain antagonistic paradigms or interpretations. The boundary between the core and the periphery of a discipline is more or less amorphous and flickering. This boundary is influenced by the socio-political conditions and the level of technological development of that time.

The geocentric paradigm of Ptolemy compelled astronomers to explain the motion of the planets in terms of complex and weird epicycles in the Middle Ages. This model was in a way inspired by the dominant world view that man was a creature of an omnipotent God and that man was at the centre of the Universe. Copernicus shattered this model by propounding another revolutionary paradigm- the heliocentric model of the solar system (the Earth and the planets revolve around the Sun). This model had a tremendous impact on the development of science in the Western world, paving the way to the triumph of Newtonian Mechanics and various scientific fields.

The rise of Modern Physics in the beginning of this century was also a breakthrough in the dominant

paradigm of physics in the 19th Century. With refined techniques of observation and experimentation, scientists found that Newtonian Mechanics and classical electrodynamics could not account for physical phenomena adequately. The prevailing world views then- the Kantian concept of absolute space and time was challenged by relativity; the continuous concept of energy was replaced by the concept of discrete quanta in quantum mechanics.

The transition from a dominant paradigm to another a scientific discipline is usually characterized by confusion, resistance and conflict. Previous conventions and rules may be destroyed. Some of the practitioners tend to defend and preserve their myths up to the point of self-delusion, regardless of the rigor and power of the new paradigm. These practitioners may ally with the political power-holders and persecute the adherents of the new model.

In our modern world, scientific research is often institutionalized and run by the government or large business corporations. It is geared to the economic, political and military arena within a nation and between various nations. The pursuit of science, in some way, has become the struggle for power, for power to control and manipulate. The power-holders define the domain of scientific inquiry and affect the process of the transmission and dissemination of scientific knowledge. The injudicious application of science makes it a dangerous game of nuclear weaponry and biochemical control.

The quantum jump of paradigms may be coupled with a revolutionary change in the world view prevalent of that age. Evolutionists hypothesize that living organisms, including man, were evolved from non-living matter. The image of man as God's creature was changed drastically to that of the progeny of the apes. Physicists have had to abandon some concepts which were so firmly held that they had probably never been questioned before. These included the concept of a time flowing constantly for all observers, the concept of the simultaneity of events, the exact specification of the motion of a particle.....Scientific knowledge is hypothetical. The world view or the image of man derived from science is by no means absolute or inviolable. It is modified or replaced according to the dominant paradigms. These dominant paradigms, in turn, shape and influence the world view at that particular period. Thus, the scientific experience of man is dynamic and historical, rather than static and ahistorical. In this way, science can become a powerful force linking theory with practice and transforming abstract thinking to concrete existence.

